



Uniqueness

Thm.

Let ϕ be a real-valued function on $R = [a, b] \times [\alpha, \beta] \subseteq \mathbb{R}^2$.

Consider the initial-value problem

$$\begin{cases} y' = \phi(x, y) \\ y(a) = C \in [\alpha, \beta]. \end{cases}$$

A differentiable function f on $[a, b]$ such that $f(a) = C$ and $\alpha \leq f(x) \leq \beta$ and

$$f'(x) = \phi(x, f(x)) \quad a \leq x \leq b$$

is called a solution for the given problem.

If $y \mapsto \phi(x, y)$ satisfies Lipschitz condition for any $a \leq x \leq b$, then the given problem has at most one solution.

Existence.

Thm.

Let f be a real-valued Continuous bounded function on $[0,1] \times \mathbb{R}$.

Then, the initial-value problem

$$\begin{cases} Y' = f(x, Y) \\ Y(0) = C \in \mathbb{R} \end{cases}$$

has a solution.

Proof